UNIT 2 GAINS FROM TRADE [*]

Structure

2.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain the meaning of gains from international trade and sources of such gains;
- identify the conditions under which two countries can gain from trading with each other;
- elucidate and illustrate how the terms of trade determine the extent to which each country specializes; and
- explain and illustrate the mutual benefits of trade.

2.1 INTRODUCTION

As we have discussed in the previous unit, international trade refers to the voluntary exchange of goods and services between different countries. Trade between countries will only occur if they consider it beneficial to their interests.

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The previous unit briefly discussed about why do countries trade? In this unit, we shall elaborate the concept of gains from international trade.

In this connection, the meaning of gains from trade will be discussed followed by sources of gains and factors determining gains from trade. Thereafter, we shall discuss the concept of the Production Possibilities Curve which shows all the various alternative combinations of the two commodities that a country can produce by completely utilising all its factors of production with the best available technology.

Subsequently, we will explain how gains from trade can be measured. Which imply how each nation gains from international trade by exchange and specialisation in the production?

Lastly, we analyse the various situations of the potential and actual gain, Free Trade vs. no trade and static and dynamic gains.

2.2 MEANING OF GAINS FROM TRADE

Gains from international trade refer to the various benefits which country derived by way of trading of goods and services with other countries. Such benefits from trade are because of international division of labour and specialisation. Countries trade with each other because trade is beneficial to all. Basic motivation of trade is the gain or benefit that nations accrue over the period. Each trading country gains when the total output increases as a result of the division of labour and specialisation. In the case of autarky or isolation, the benefits of the international division of labour do not flow between nations.

Malthus held that ‘the gain from trade consisted of the increased value which results from exchanging what is wanted less for what is wanted more, and that 'international trade, by giving us commodities much better suited to our wants and tastes than those which had been sent away, has decidedly increased the exchangeable value of our possessions, our means of enjoyment, and our wealth’.

In simple words, gains from trade refers to additional production and consumption effects that countries can achieve by way of entering trade with each other. To sum-up, gains from international trade are of two types–i) gain from exchange and ii) gain from specialisation in production. Approaches preferred by various economists will be discussed in subsequent sections.

2.3 SOURCES OF GAINS

As per classical theory, major source of gains from international trade arises out of specialisation based on the principle of comparative cost advantage. While pointing out the 'division of labour’, Adam Smith has held that it is limited by the size of the market. Upon expansion of the market size because of international trade, the scope for large-scale production also increases which in turn increases the scope for complex division of labour and specialisation. In short, international trade leads to enlargement of the market which further leads to specialisation and division of labour. This process further results in an increase in output per unit of input.

The comparative cost theory exhibits increased world production as gains from international trade in the real-world scenario. Each trading country gains by way of getting relatively more, better and cheaper goods. In the entire process no country lose by having less than what they needed.

In this way we can say that gains from trade are results of specialization in production from the division of labour, economies of scale, and agglomeration along with relative availability of factor resources.

When there is free trade, goods and services produced all over the world are available to people everywhere. In other words, international trade makes available to the people of a country, a galaxy of goods and services at the most competitive prices. A country may not have the factor endowments or technological capability to produce certain goods. If there is no trade with other countries, it will have to do without such goods but through international trade, it can procure them.

The gains from international trade may be summed up as follows:

i) Expansion of the size of the market

ii) Division of labour

iii) Gains from specialisation

iv) Gains from increased product variety

v) Gains from competition

vi) Gains from increased economies of scale

vii) Productivity gains.

2.4 FACTORS DETERMINING SIZE OF GAINS

Differences in cost ratios: if country A has a comparative advantage in the production of wheat and country B has a comparative advantage in the production of cotton, both countries will gain from trade. The size of the gain will depend on the cost of production of each commodity in both countries. The gain from international trade depends upon the cost ratios of differences in comparative cost ratios in the two trading countries. The larger the difference between exchange rate and cost of production the larger the gains from trade and vice versa.

1) **Reciprocal demand:** The reciprocal demand means the relative strength and elasticity of demand of one country for the product of the other in exchange for its product.

2) **Level of income:** The level of money income of a country is another factor which determines the gains and the share of trade. A country whose goods have constant demand in other countries will have a high level of money income.

3) **Terms of trade (also called "trading price"):** It is the most important factor which determines the gains from trade. The international terms of trade refer to the rate at which one commodity of a country is exchanged for another commodity of the other country. If the cost ratio and terms of trade are closer to each other, more will be the gains from the trade of the participating countries.

4) **Productive efficiency:** An increase in the productive efficiency of a country also determines its gains from trade. It lowers the costs of production and prices of goods in the home country. As a result, the other country gains by importing cheap goods and its terms of trade improve.

5) **Nature of commodities exported:** Another factor is the nature of commodities exported by a country. A country that exports mainly primary products has unfavourable terms of trade. Consequently, it gains from trade will be smaller. On the contrary, a country exporting manufactured goods has favourable terms of trade and its gain from trade will be larger.

6) **Technological conditions:** In a country which is technologically advanced and has an abundance of capital, its volume of foreign trade will be large and so it will gain from international trade. On the other hand, if a country is technologically backward with abundant labour, its volume of foreign trade will be small and so will be gains from trade.

7) **Size of the country:** If a country is small, it is relatively easy for them to specialize in the production of one commodity and export the surplus production to a large country and can get more gains from international trade. Whereas if a country is large, then they must specialise in more than one good because the excess production of only one commodity cannot be exported fully to a small-sized country as the demand for the commodity will reduce very frequently. So, the smaller the size of the country, the larger the gains from trade.

8) **Factor availability:** International trade is based on specialization and a country specializes depending upon the availability of factors of production. It will increase the domestic cost ratios and thereby the gains from trade.

9) **Productive Efficiency:** An increase in the productive efficiency of a country also determines its gains from trade. A highly efficient country can keep the cost of production and price of the goods lower than other countries.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Why do countries trade with each other? What are the gains from trade?

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2) Examine the main source of gains from trade?

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3) How do the gains from trade help an economy grow?

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2.5 PRODUCTION POSSIBILITIES CURVE IN INTERNATIONAL TRADE

Production Possibilities Curve (also known as production frontier, transformation curve, product substitution curve or an opportunity cost curve) shows all the various alternative combinations of the two commodities that a nation can produce by fully utilizing all its factors of production with the best available technology.

The slope of the production possibilities curve refers to the marginal rate of transformation (MRT) or the amount of a commodity that the nation must give up to get one additional unit of the second commodity.

If the nation faces constant costs or MRT, then its production possibilities curve is a straight line with an (absolute) slope equal to the constant opportunity costs or MRT and the relative commodity price in the nation.

The country lacks the capacity beyond the limit specified by the production possibility curve. It shows the production frontier of the country.

Producing output beyond the production frontier is not possible. If the output of the two or one of the two commodities is below the production frontier, that indicates unemployment or excess capacity. To be more specific, consider a situation when the production of Y commodity is reduced to produce more units of X commodity, it signifies that Y has been transformed or substituted into X-commodity. Therefore, it is called an opportunity cost curve, transformation curve or product substitution curve.

Example: When all the available resources are employed in production of only one commodity say commodity X, commodity X can be produced at maximum possible level without any production of another commodity say Y. On the opposite, if all the resources are utilised in the production of Y, the country will be able to produce maximum quantity of Y commodity with no output of X commodity. Between these two extreme situations, there can be various production possibilities involving more or fewer quantities of the two commodities. If the production of X is to be increased, there will be a diversion of resources from the production of Y to the production of X, resulting in reduced production of Y.

MRT is the ratio of a change in the quantity of commodity Y to a change in the quantity of X commodity.

$MRT_{xy} = \delta y/\delta x$

Since additional production of X involves reduced output of Y, the MRT is negative. It signifies that the production possibility curve or opportunity cost curve slopes negatively, or it slopes downwards from left to right.

The MRT_{xy} can be expressed also as a ratio of the marginal cost of X to the marginal cost of Y.

It can be determined as below:

$\delta C = MC_x.\ \delta x + MC_y.\ \delta y$

Where δC = Change in cost, δx = Change in the quantity of X commodity, δy = Change in the quantity of Y commodity. MC and MC are the marginal costs of X and Y commodities respectively.

If

$\delta C = 0$

$MC_x.\ \delta x + MC_y.\ \delta y = 0$

$MC_x.\ \delta x = -\ MC_y.\ \delta y$

$$\frac{MC_x}{MC_y} = -\frac{\delta_y}{\delta_x}$$

or $MRT_{xy} = -\frac{\delta_y}{\delta_x} = \frac{MC_x}{MC_y}$

If the production is governed by constant returns, the MC_x relative to MC_y remains unchanged or MRT_{xy} remains the same. It means the slope of the production possibility curve or opportunity cost curve is the same and it is a negatively sloping straight line. If the production is governed by diminishing returns, MC_X rises relative to the MC_y. It signifies that the slope or MRT_{xy} increases. In such a situation, the opportunity cost curve is a negatively sloping concave curve to the origin. If the production is governed by increasing returns, the MC_X decreases relative to the MC_y. The slope or MRT_{xy} decreases. In this case, the opportunity cost curve is a negatively sloping convex curve to the origin. These cases are depicted in Fig. 2.1 (a), 2.1 (b) and 2.1(c) respectively.

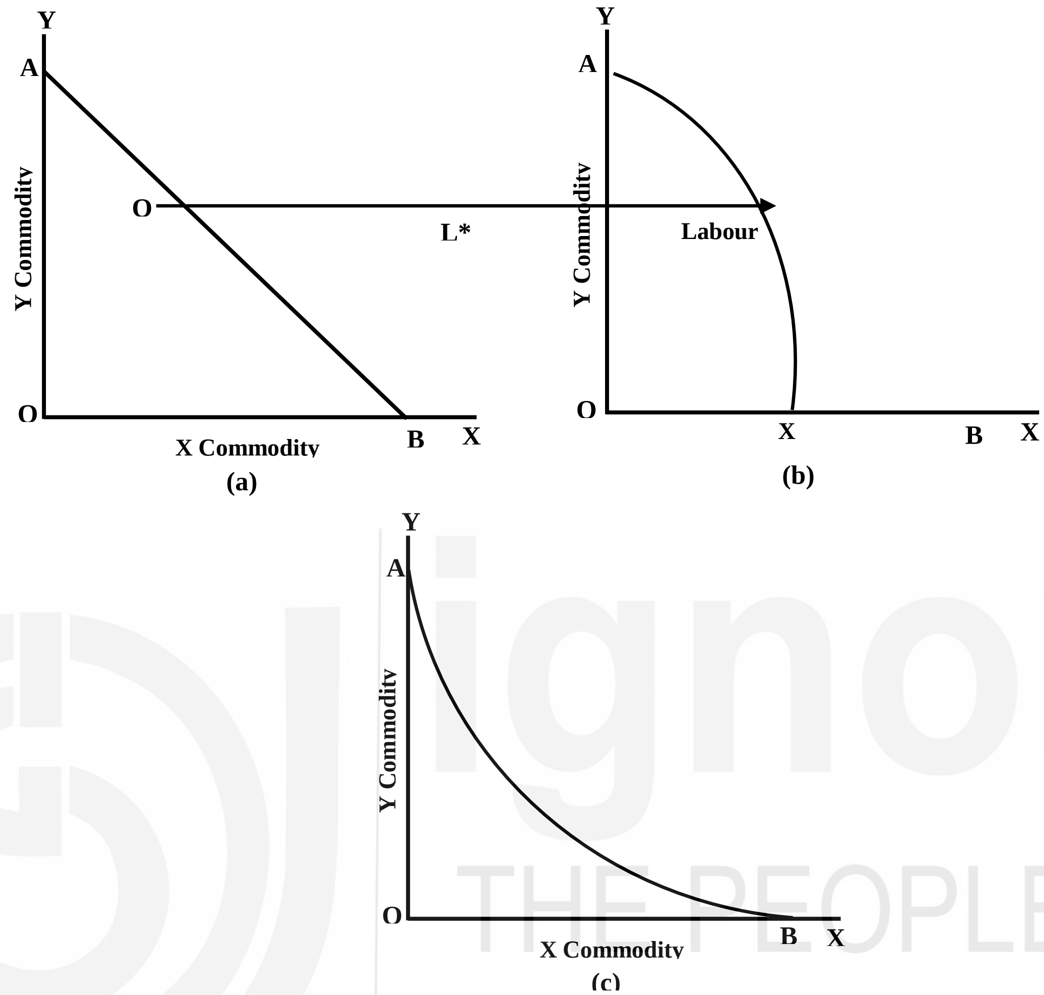

Figure 2.1(a), 2.1(b) and 2.1 (c)

The opportunity cost curve simply indicates the alternative production possibilities. It does not show what combinations of the two commodities will be produced.

In Fig. 2.1, AB is the production possibility curve or the opportunity cost curve. In Fig. 2.1 (a), the opportunity cost curve AB is the negatively sloping straight line. Along this curve, MRTxy= $-\delta y/\delta x$ = MCx/MCy remains unchanged due to constant opportunity cost conditions (constant return)

In Fig. 2.1 (b), the opportunity cost curve AB is negatively concave. Along it, MRTxy = $-\delta y/\delta x$ = MCx/MCy increasing opportunity cost condition (diminishing returns).

In Fig. 2.1 (c), the opportunity cost curve AB is a negatively sloping convex curve to the origin on account of decreasing opportunity cost condition (increasing returns).

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Why is the production possibilities curve of each nation usually different?

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2) Is trade still possible between two nations if they have identical production possibilities curves?

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3) Under what conditions can no trade take place between two nations with different production possibility curves?

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4) What happens if two nations are identical in every respect?

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2.6 MEASUREMENT OF GAINS FROM TRADE

Jacob Viner pointed out that the gains from trade were measured by the classical economists in terms of:

i) An increase in national income,

ii) Differences in comparative costs, and

iii) Terms of trade.

Some approaches to the concept of gains from trade and their measurement are discussed below:

1) **Adam Smith's Approach:**

According to Adam Smith, international trade helps to improve the productive capacity of trading nations. It increases value of the product as excess of a product in domestic market can be exported. A nation can also import products which are high in demand in domestic market. Thus, gains from international trade are in form of maximum welfare through maximum possible export earnings.

A country reaches the point of optimum allocation as each country specializes in production of a commodity in which it has cost advantage. Specialization results in economies of scale i.e., reduced cost of the product.

2) **Ricardo-Malthus Approach:**

In unit 1, we have discussed about Comparative advantage theory. You can recall that Ricardo asserted that principle of comparative costs in international trade results in saving costs and resources.In the words of David Ricardo, “The advantage to both places is not that they have any increase in value but with the same amount of value they are both able to consume and enjoy an increased quantity of commodities."

Malthus had expressed in this regard views similar to those of Adam Smith. The gain from trade, according to him, consists of "the increased value, which results from exchanging what is wanted less for what is wanted more." The international exchange on this basis increases the "exchangeable value of our possession, our means of enjoyment and our wealth."

The Ricardo-Malthus approach to gains from trade was illustrated by Ronald Findlay in Fig. 2.2.

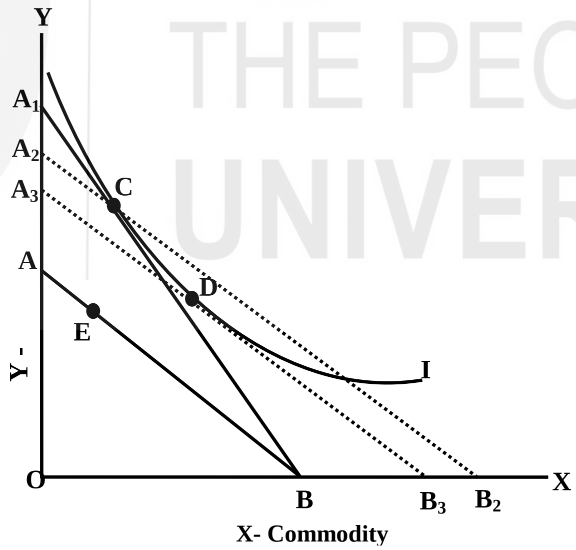

Figure 2.2: The Ricardo-Malthus approach to gains from trade

In Fig. 2.2., AB is the production possibility curve under constant cost conditions before the trade. Suppose the production takes place at E. If this country enters into trade, the international exchange ratio line shifts to A_1B and production of two commodities X and Y now takes place at C. The production at point C will be possible if the labour input increases an extent that the production possibility curve shifts to A_2B_2. The gain from trade will be measured by BB_2/OB.

Malthus criticized this measure of gain from trade as exaggerated. According to him, if the production possibility curve shifts to A_2B_2, point C cannot be the point of equilibrium. The relative prices along A_2B_2 are more favourable to the export good X than along the line A_1B. Some points to the right of C rather than C itself would be preferable to the community. Hence the gain from trade along line A_1B cannot be measured by an increase in the input of labour in the ratio BB_2/OB. It tends only to overstate the gain from trade.

Ronald Findlay attempted a modification over the Ricardian measure of gain from trade by introducing in this analysis the community indifference curve. Given the community indifference curve I, the equilibrium does not take place at the Ricardian trade equilibrium position C but at D where the production possibility curve A_3B_3 became tangent to the community indifference curve I.

At point D, everyone in the community is better off than at C. The gain from trade in this situation is BB_3/OB rather than BB_2/OB. The measure of gain from trade BB_3/OB vindicates the Malthusian criticism that the Ricardian measure of gain from trade was an over-estimation.

3) **J.S. Mill's Approach:**

Ricardian approach did not explain the distribution of gains from trade among the trading countries. J.S. Mill in his approach attempted to analyse both the gains from trade and the distribution thereof among the trading countries. He stressed on the concept of reciprocal demand.

According to him, reciprocal demand determines terms of trade, which is a ratio of quantity imported to the quantity exported by a given country. The terms of trade determine the distribution of gains of trade between the trading partners.

Suppose in country A, 1units of labour can produce 10 units of X and 10 units of Y so that the domestic exchange ratio in country A is: 1 unit of X = 1 unit of Y. In country B, 2 units of labour can produce 12 units of X and 18 units of Y so the domestic exchange ratio in this country is: 1 unit of X = 1.5 units of Y. The domestic exchange ratios set the limits within which the actual exchange ratio or terms of trade will get determined.

The reciprocal demand or the strength of the elasticity of demand of the two trading countries for the products of each other will decide the actual rate of exchange of two commodities. If A's demand for commodity Y is less elastic, the terms of trade will be closer to its domestic exchange ratio: 1 unit of X = 1 unit of Y. In this case, the terms of trade will be favourable for country B and against country A.